# Random Forests, Decision Trees, and Categorical Predictors: The "Absent Levels" Problem

Erik M. Painter

Department of Mathematics & Statistics

STA 685 – Statistical Machine Learning

12/05/2025



## Table of Contents

Background & Goals……3

Introduction & Problem……4

Data Description……6

Statistical Analysis……7

Conclusions……9

Further Considerations……...11

References……13

## 1. Background & Research Goals

Since their introduction in Breiman (2001), random forests have been among the most widely used decision tree-based methods in machine learning algorithms due to their strong predictive accuracy and interpretability. One main advantage decision tree-based methods (such as random forests) have been their ability to effectively handle categorical predictors without having to first transform them. However, the following analysis will show how this capability can lead to an inherent "absent levels" problem. This problem occurs whenever an **indeterminacy over how to handle an observation that has reached a categorical split which was determined when the observation in question's level was "absent" during training** – this issue can arise in 3 distinct ways:

1. Levels are present in the population, but due to sampling variability, are absent in the training set
2. The levels are present in the training set but, due to bagging, are absent in an individual tree's bootstrapped sample of the training set
3. The levels are present in an individual tree's training set but, due to a series of earlier node splits, are absent in certain branches of the tree

These occurrences result in situations where observations with absent levels are unsure of how to proceed further down the tree – an issue that has been overlooked in theoretical literature and the software implementation of these methods. In this analysis we examine how overlooking this problem can systematically bias a model, dramatically altering a model's performance in practice. There is no theoretical framework yet established to deal with the absent level problem. However, simple heuristics employed by Au (2018) can be used to help mitigate the effects of the absent levels problem.

The effects differ based on the type of task (regression/classification), data structure (high cardinality of categorical variables, class balance, etc.), range of our response variables, etc. I chose to focus on a specific case:

**A regression tree used to predict the sale price of a home in Ames, Iowa**

## 2. Introduction & Problem

In random forests, each tree is trained on a bootstrap sample of the original data. When a categorical variable contains levels that do not appear in the bootstrap sample, any split involving this categorical variable lacks the necessary information to send OOB observations to the correct daughter node. This is the absent-levels problem.

The absent levels problem most commonly occurs in the following situations:

i. small sample settings with high cardinality categorical variables

ii. bootstrap samples that fail to capture all category levels

To understand the problem, we must first discuss how unordered categorical variables are split in a regression tree:

- $p$: predictor; $s_p$ – subset of predictor p's domain that gets sent to $N_L$ and $N_R$
- Optimal pair is pair which minimizes the measure of node impurity – in the case of regression, this is **squared error loss**

$$(p^*, \mathcal{S}_{p^*}^*) = \underset{(p,\mathcal{S}_p)}{\arg\min} \left( \sum_{n \in \mathcal{N}_{\mathcal{L}}(p,\mathcal{S}_p)} [y_n - \hat{c}(\mathcal{N}_{\mathcal{L}}(p,\mathcal{S}_p))]^2 + \sum_{n \in \mathcal{N}_{\mathcal{R}}(p,\mathcal{S}_p)} [y_n - \hat{c}(\mathcal{N}_{\mathcal{R}}(p,\mathcal{S}_p))]^2 \right)$$

Predictions in nodes $N_L$ and $N_R$ are the **means** of the training observations within these nodes

To split on an unordered categorical predictor, we must apply a procedure introduced by Fisher (1958)

- Training observations in $N_M$ are used to calculate the mean response within each of p's unordered levels

$$\gamma_p(q) = \text{ave}(y_n \mid n \in \mathcal{N}_{\mathcal{M}} \text{ and } x_{np} = q), \quad q \in \mathcal{Q}.$$

- Means are then assigned to every training observation in $N_M$ according to its observed level for predictor p

$$\tilde{x}_{np} = \gamma_p(x_{np}), \quad n \in \mathcal{N}_{\mathcal{M}}.$$

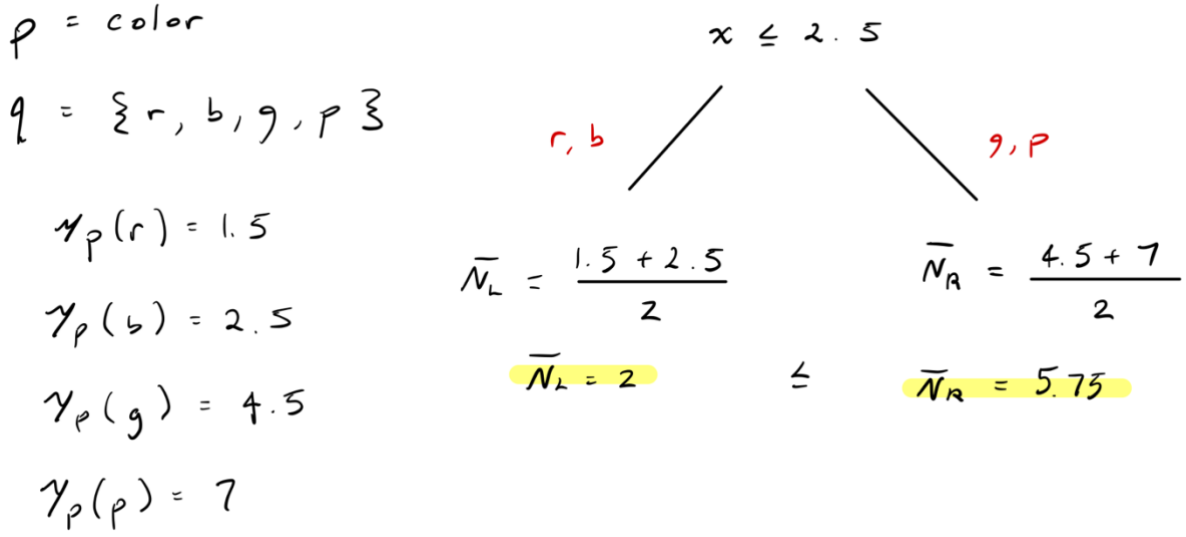
- Optimal "pseudo" splitting criterion is determined by doing an ordered split on these numeric pseudo values, which minimizes the squared error loss impurity measure

$$\mathcal{N}_{\mathcal{L}}(p, \tilde{\mathcal{S}}_p^*) = \{n \in \mathcal{N}_{\mathcal{M}} : \tilde{x}_{np} \leq \tilde{s}_p^*\} \quad \text{and} \quad \mathcal{N}_{\mathcal{R}}(p, \tilde{\mathcal{S}}_p^*) = \{n \in \mathcal{N}_{\mathcal{M}} : \tilde{x}_{np} > \tilde{s}_p^*\}.$$

**What are the implications?**

The prediction is given by the mean of the training responses that are in a node – therefore, since our prediction is a weighted average across the means of the present levels within that node, it follows that:

- $N_L$ will **always** give a prediction that **is smaller** than the right daughter node, $N_R$
    - When splitting on a categorical predictor p in a regression tree that uses a squared node impurity measure
- **randomForest** R package employs this methodology




This method, along with its implementation in **randomForest**, applies 'zero imputation' to the absent levels:

$$\gamma_p(q) = \begin{cases} \text{ave}(y_n \mid n \in \mathcal{N}_{\mathcal{M}} \text{ and } x_{np} = q) & \text{if } q \in \mathcal{Q}_{\mathcal{P}} \\ 0 & \text{if } q \in \mathcal{Q}_{\mathcal{A}} \end{cases},$$

In the case when our response is strictly positive – Sales Price ($) - the OOB observations with the absent level problem is **always** be sent to the left daughter node $N_L$, as $sp^* > 0$.

Due to systematic differences that exist among the two daughter nodes, the arbitrary decision of sending the absent level to $N_L$ (or $N_R$) can significantly impact the models' final predictions.

**3. Data Description**

NAME: AmesHousing.csv

TYPE: Population

SIZE: 2930 observations, 82 variables

DESCRIPTIVE ABSTRACT: Data set contains information from the Ames Assessor's Office used in computing assessed values for individual residential properties sold in Ames, IA from 2006 to 2010.

- Alternative to the Boston Housing Data Set

SOURCES: Ames, Iowa Assessor's Office

Obtained this data set from the **modeldata** R package (De Cock 2011)

- Data set is cleaned up & transformed

**Predictors**: Mix of numeric & categorical variables (ordinal/nominal)

- Numeric: Lot Area, Basement SF, 1st Floor SF, 2nd Floor SF, Garage Area, Pool Size, Deck Area, etc.
- Categorical: Lot Shape, Land Slope, Neighborhood, Overall Condition, Roof Style, etc.

**Response**: Sale Price

**High cardinality categorical variables such as *Neighborhood* create an environment for the absent levels to exist**

| MS_SubClass | MS_Zoning | Lot_Frontage | Lot_Area | Street |
|---|---|---|---|---|
| One_Story_1946_and_Newer_All_Styles | Residential_Low_Density | 141 | 31770 | Pave |
| One_Story_1946_and_Newer_All_Styles | Residential_High_Density | 80 | 11622 | Pave |
| One_Story_1946_and_Newer_All_Styles | Residential_Low_Density | 81 | 14267 | Pave |
| One_Story_1946_and_Newer_All_Styles | Residential_Low_Density | 93 | 11160 | Pave |
| Two_Story_1946_and_Newer | Residential_Low_Density | 74 | 13830 | Pave |
| Two_Story_1946_and_Newer | Residential_Low_Density | 78 | 9978 | Pave |

## 4. Statistical Analysis

This section summarizes the key analysis stages implemented to replicate the methodology of Au (2018).

Since random forest behavior varies across different bootstrap samples, each component of the analysis was run across 100 replications, with 200 trees fit in each. This was mainly done for computation reasons, as the results did not differ much as increased the tree or replication count. Using these replications allowed me to stabilize the comparisons across heuristics, allowing us to obtain a more robust comparison across the different heuristics.

The data incudes several high-cardinality categorical predictors. However, since the Ames Housing dataset is quite large, we had to down sample to 200 observations per replication to intentionally create an environment where the absent-levels problem would occur frequently.

### a. Computing OOB absence proportions

Using **randomForest** in bag matrix, for each tree within each replication we identified:

- In-bag observations
- Out-of-bag observations
- Categorical levels absent from the training data
- For each observation, we computed:

**An OOB observation exhibits an "absent level" in each tree if it has a categorical value that *did not appear at all* in the tree's training data. For each observation $i$, we computed:**

$$OOB_i = \frac{1}{B}\sum_{b=1}^{B} I(\text{observation } i \text{ has an absent level in tree } b)$$

Distribution results show OOB absence proportions for each OOB observation across all 100 replications (Au, 2018)

### b. Simulating Heuristic Predictions

Since modifying the internal C code of **randomForest** is not possible, we had to simulate the effects in Section 3 of Au (2018). For each heuristic (Left, Right, Stop, Majority, Random, DBI), predicted values were adjusted as a function of the OOB absence proportions.

- **Left:** underprediction
- **Right:** overprediction
- **Stop:** shrinkage toward parent node mean or overall mean
- **Majority/Random:** low directional bias, increased noise

- **DBI:** convex mixture toward global mean/probability

These simulated adjustments replicate the fundamental behaviors described in Section 3 of Au (2018).

**c. Pairwise Differences Across Replications**

We reproduced Figure 2 from Au (2018), which assesses how strongly the heuristics differ from one another. For each replication *r*, each heuristic *h*, and each observation *n*, we computed the adjusted OOB prediction:

$$\hat{y}_{nr}^{(\mathrm{h})}$$

For every pair of heuristics $(h_1, h_2)$, we computed within-replication differences:

$$\hat{y}_{nr}^{(h_1)} - \hat{y}_{nr}^{(h_2)}$$

These differences were then grouped into bins according to the OOB absence proportion of each observation (0–1 on a 0.05 grid). For each bin, we computed:

- **Mean difference**
- **2.5% and 97.5% quantiles**, which form a **95% interval**

The **mean difference** in a bin represents the average difference between two heuristics' predictions for observations that experienced a similar level of absent-level frequency. A positive mean indicates that $h_1$tends to predict larger values than $h_2$; a negative mean indicates the opposite. These paired panels clearly show that differences across heuristics grow larger as the absent levels become more frequent.

**d. RMSE Plots**

We evaluated the overall performance of each heuristic $h \in H$ within an experimental replication $r$ in terms of its RMSE.

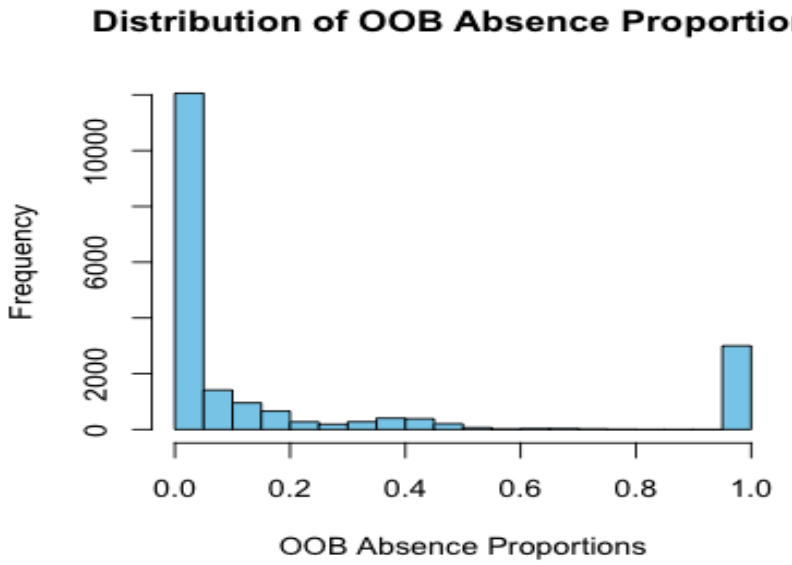
$$\mathrm{RMSE}_r^{(h)} = \sqrt{\frac{1}{N}\sum_{n=1}^{N}\left(y_n - \hat{y}_{nr}^{(h)}\right)^2}.$$

We reproduce Figure 3 from Au (2018), which displays each heuristics marginal distribution of RMSE's across the 100 replications.

## 5. Conclusions

Following the statistical analysis from part 4, we will highlight the key components and findings of the analysis.

Below is the distribution for the OOB absence proportions across all 200 trees and 100 replications. A heavy right-skew exists, with many observations having near 100% absent levels. We can see that most of the observations in this example had the absent levels problem occur in less than 5% of their OOB trees.




I decided to quantify how the different heuristics influence predictions, let $\hat{y}_{nr}^{(h)}$ denote the OOB prediction produced by heuristic $h$, for observation $n$, in replication $r$. For each replication, and for every ordered pair of heuristics $h_1, h_2 \in H$, we computed the within-replication difference

$$\hat{y}_{nr}^{(h_1)} - \hat{y}_{nr}^{(h_2)}.$$

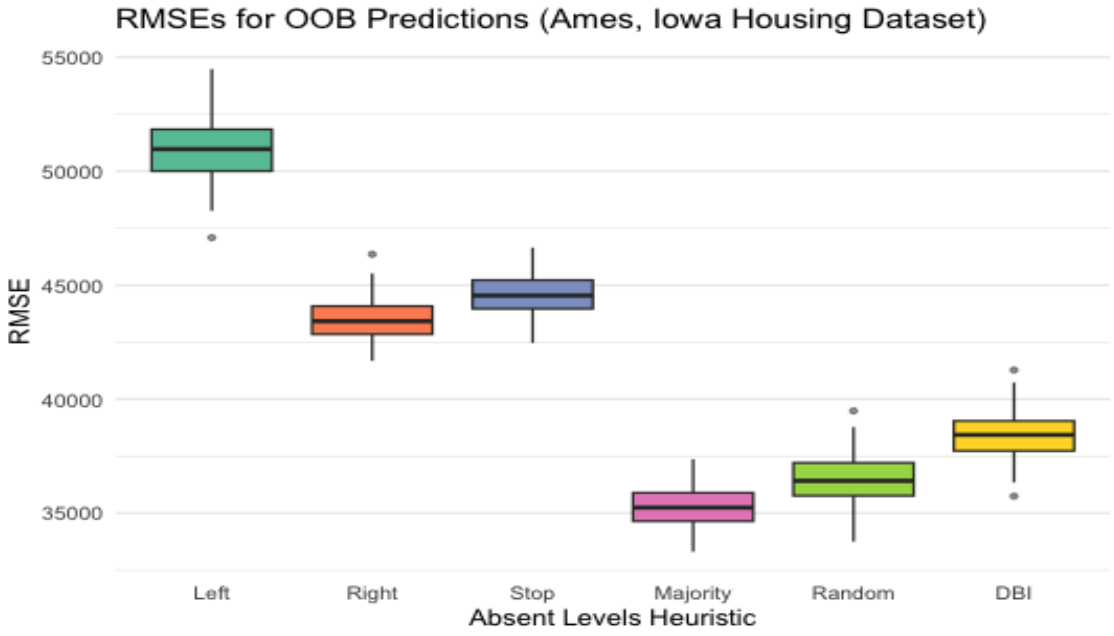
These differences were then grouped according to the **OOB absence proportion bin** of each observation (e.g., 0–5%, 5–10%, etc.). Within each bin, we summarized the pairwise differences by taking the mean and the middle 95% of values across all 100 replications. Thus, each plotted point represents the **average difference among all observations whose absence proportion falls within that bin,** unlike Au (2018) which shows difference and 95% confidence interval for each observation.

We observe that these differences become more pronounced for bins corresponding to larger OOB absence proportions: both the mean difference and the width of the intervals tend to increase as the absence proportion grows. This behavior is consistent with intuition, since

heuristics exert greater influence precisely when absent levels occur more frequently; binning the data reveals this trend cleanly across the full range of absence proportions.

Finally, we can evaluate the overall performance of each heuristic, within an experimental replication r, in terms of its RMSE. Each boxplot represents the marginal distribution of the RMSE's across the 100 experimental replications. A few key takeaways emerge:

- **Left** heuristic performed the worst - tendency to severely underpredict
- Sending absent levels to the $N_L$ (particular to this dataset) substantially underperformed relative to other heuristics
- **Majority** heuristic achieved the lowest RMSE overall, with Random and DBI performing similarly
- **Stop** performed the worst of all missing data heuristics.




It is important to emphasize that these results are specific to this datasets structure. Different datasets may favor different heuristics; however, these findings show that the choice of absent level handling can have a strong influence on model accuracy.

## 6. Further Considerations

### i. Classification Setting

Although the main analysis in this project focuses on the regression case, the absent levels phenomenon applies almost identically to binary classification random forests. The same issue exists.

For a binary classification tree using Gini Index as the node impurity measure, the splitting behavior on a categorical predictor is characterized in the same way as regression – finding optimal pseudo numeric split. The splitting criterion is not the ordered means, rather it is the ordered proportions of observations belonging to class k = 1.

$$\gamma_p(q) = \frac{|\{n \in \mathcal{N}_{\mathcal{M}} : x_{np} = q \text{ and } y_n = 1\}|}{|\{n \in \mathcal{N}_{\mathcal{M}} : x_{np} = q\}|}, \quad q \in \mathcal{Q},$$

Those with proportions that are less than $sp^*$ get sent to the left daughter node, $N_L$, and those that are greater get sent to the right daughter node, $N_R$. This is **directly** analogous to the regression case where levels are ordered, and subsequently split, by their mean response.

Since each daughter nodes predicted class probability is a weighted average of the values within it, the following always hold true under Gini splitting:

**The left daughter node $N_L$ always has a lower probability of class $k = 1$ than the right daughter node $N_R$.**

In other words, observations with the absent level problem are biased towards the minority class. Note that the implications are sensitive towards to the actual ordering of the response classes.

### ii. Recent Developments (2024)

The 2024 paper *Lost in the Forest: Encoding Categorical Variables and the Absent Levels Problem* directly addresses the systematic bias introduced by absent levels in random forests. It proposes a more structured solution using a PCO-encoding method, which maps categorical levels into a continuous coordinate space based on their similarity to one another rather than their relationship to the response variable. This approach provides a robust way to mitigate the effects of absent levels and avoid the biases inherent in traditional, target-based encodings.

Technically, this is done by computing a similarity (or dissimilarity) matrix between levels and using principal coordinates analysis to assign each level a position $z_q$ so that distances, $\| z_q - z_{q'} \|$, reflect how alike the levels are. An absent level observation is then placed near the levels it most closely resembles, instead of being assigned an extreme class-probability value such as

$\gamma_p(q) = 0$, which in the **randomForest** package forces it into the left child and introduces systematic bias.

A helpful way to visualize this is to imagine each level as a city on a map: PCO builds the map based on the distances between cities, and when a new city (an absent level) appears, it is placed near the cities it resembles rather than being dropped arbitrarily at the bottom of the map. By grounding absent levels in their true similarity structure, PCO avoids the distortions caused by target-based encodings and produces more stable, unbiased, and interpretable predictions.

## 7. References

Smith, H.L., Biggs, P.J., French, N.P. *et al.* Lost in the Forest: Encoding categorical variables and the absent levels problem. *Data Min Knowl Disc* **38**, 1889–1908 (2024). https://doi.org/10.1007/s10618-024-01019-w

Au, Timothy C. "Random Forests, Decision Trees, and Categorical Predictors: The 'Absent Levels' Problem." *Journal of Machine Learning Research*, vol. 19, 2018, pp. 1-30. Google Research Publications. research.google/pubs/random-forests-decision-trees-and-categorical-predictors-the-absent-levels-problem/.

Breiman, L. Random Forests. *Machine Learning* **45**, 5–32 (2001). https://doi.org/10.1023/A:1010933404324

De Cock, D. 2011. "Ames, Iowa: Alternative to the Boston Housing Data as an End of Semester Regression Project." *Journal of Statistics Education* 19 (3).